

GYM MANAGEMENT & AI PLATFORM

System Requirements & Functional Feature Specification

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This document details the refined, production-ready core functional capabilities of the Gym Management Platform. The features are modularly distributed across customer, trainer, administrator, and artificial intelligence boundaries to support a scalable, modern microservices architecture.

1. Customer Capabilities

- **Real-Time Occupancy Tracking:** Customers can view the live count of active individuals inside the gym facility, enabling optimal training session planning and crowd avoidance.
- **Comprehensive Progress Analytics:** Personalized dashboard to log and track physical metrics over time, including customized nutrition programs, weight fluctuations, and detailed InBody body composition metrics.
- **Social Community Threads:** A built-in community bulletin where members can post threads to sync schedules, find workout partners, foster companionship, and optimize communication.
- **Flexible Booking & Subscriptions:** Self-service module allowing clients to seamlessly book, subscribe to, or reschedule physical sessions, group classes, and dedicated personal trainer programs.
- **Membership Benefit Tracker:** A transparent ledger detailing plan benefits, tracking available freezing/paused days, remaining InBody scan tokens, individual PT session counts, and days left in the current membership cycle.
- **Direct Chat & Messaging:** Unified internal messaging system enabling instant peer-to-peer customer chats or secure direct channels with designated personal trainers.
- **E-Commerce Marketplace:** In-app storefront where customers can purchase premium gym products such as nutritional supplements, fitness gear, custom merchandise, or bundled program offers.
- **Integrated Secure Payments:** Seamless payment gateway handling transactions across the platform, covering flexible membership renewals, trainer coaching fees, and e-commerce checkouts.

2. Trainer Capabilities

- **Professional Portfolio Management:** Dedicated space for trainers to build their internal CV by uploading certified credentials, professional achievements, specializations, and career bios.
- **Client Progress Monitoring:** Granular analytical view of all assigned clients, allowing the trainer to review historical weight entries, body composition, and adherence metrics.
- **Client Communication Channel:** Immediate and organized chat access to message active clients, deliver motivation, and answer technical inquiries.

- **Custom Routine & Program Builder:** Interactive tool designed for trainers to compile, schedule, and push bespoke nutrition blueprints and structured weightlifting regimes directly to clients.

3. Administrator Capabilities

- **Identity & Access Management (IAM):** Core provisioning capabilities enabling administrators to securely create, configure, update, or deactivate dedicated accounts for both customers and trainers.
- **Global Announcements & Broadcasts:** Comprehensive marketing and news engine allowing admins to post global events, target announcements for new staff onboardings, launch upcoming group classes, or feature promotional product drops.

4. Intelligent AI Features

- **Autonomous Workout & Nutrition Generation:** AI-driven consulting engine that processes localized biometric data and specific client goals to dynamically output optimized nutrition meal plans and tailored weight training splits.
- **Computer Vision Motion & Form Correction:** Interactive visual evaluation module utilizing edge computing or video analysis to detect body positions and motion pathways in real-time, offering instant modifications for safe posture.
- **Smart Schedule Optimization:** The user provides their weekly free days and optimal time frames; the AI algorithm builds a structured, efficient routine. Upon user satisfaction, the schedule triggers an automated workflow to the corresponding trainer for formal acceptance or rejection.
- **Automated Conversational Assistant:** Advanced context-aware chatbot capable of detailing application features, answering common FAQs, and directly automating functional client requests (e.g., freezing a membership or looking up point balances) via backend service routing.

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